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A List of gender bias metrics, as presented in Table 1

Many of the items in this list do not aim to describe a specific metric, but rather describe a family of metrics with similar characteristics and requirements.

A.1 Extrinsic Performance

This class of extrinsic metrics measures how a model's performance is affected by gender. This is computed with respect to particular gold labels and there is a clear defintion of harm derived from the specific performance metric measured, for instance F1, True Positive Rate (TPR), False Positive Rate (FPR), BLEU score for translation tasks, etc.

1. **Gap (Label):** Measures the difference in some performance metric between Female and Male examples, in a specific class. The performance gap can be computed as the difference or the quotient between two performance metrics on two protected group. For example, in Bias in Bios (De-Arteaga et al., 2019) one can measure the TPR gap between female teachers and male teachers. The gaps per class can be summed, or the correlation with the percentage of women in the particular class can be measured.

2. **Gap (Stereo):** Measures the difference in some performance metric between pro-stereotypical (and/or non-stereotypical) and anti-stereotypical (and/or non-stereotypical) instances. A biased model will have better performance on pro-stereotypical instances. This can be measured across the whole dataset or per gender / class.

3. **Gap (Gender):** Measure the difference in some performance metric between male examples and female examples, across the entire dataset. In cases of non-binary gender datasets (Cao and Daumé III, 2021), the gap can be calculated to measure the difference between text that is trans-inclusive versus text that is trans-exclusive. Another option is to measure the difference in performance before and after removing various aspects of gender from the text.

A.2 Extrinsic Prediction

This class is also extrinsic as it measures model predictions, but the bias is not computed with respect to some gold labels. Instead, the bias is measured by the effect of gender on the predictions of the model.

1. **% or # of answer changes:** The number or percentage that the prediction changed when the gender of the example changed. To measure this, each example should have a counterpart example of the opposite gender. This difference can be measured with respect to the number of females or males in the specific label, for instance with relation to occupation statistics.

2. **% or # that model prefers stereotype:** Quantifies how much the model tends to go

for the stereotypical option, for instance predicting that a “she” pronoun refers to a nurse in a coreference resolution task. This can also be measured as a correlation with the number of females or males in the label, which can be thought of as the “strength” of the stereotype.

3. **Pred gap:** The raw probabilities or some function of them are measured, and the gap is measured as the prediction gap between male and female predictions. This can be measured across the whole dataset or per label at other cases.

4. **LM prediction on target words**: This metric relates to the specific predictions of a pretrained LM, such as a masked LM. The prediction of the LM is calculated for a specific text or on a specific target word of interest. These probabilities are then used to measure the bias of the model. We did not include this metric category in the “Pred gap” category because it can be measured on a much larger number of datasets. For example, for the masked sentence: “The programmer said that <mask> would finish the work tomorrow”, we might measure the relation between $p(< mask >= he|sentence)$ and $p(< mask >= she|sentence)$. Although somewhat similar in idea to the previously described metric “pred gap”, it is presented as a separate metric since it can be computed on a wider range of datasets. The strategy for calculating a number quantifying bias from the raw probabilities varies in different papers. For example, Kurita et al. (2019); Nangia et al. (2020a); Bordia and Bowman (2019); Nadeem et al. (2021) all use different formulations.

A.3 Intrinsic

This class measures bias on the hidden model representations, and is not directly related to any downstream task.

1. **WEAT:** The Word Embedding Association Test (Caliskan et al., 2017) was proposed as a way to quantify bias in static word embeddings. While we consider only bias metrics that can be applied in contextualized settings, we describe WEAT here as it is popular and has been adapted to contextualized settings. To compute WEAT, one defines a set of target words X, Y (e.g., programmer, engineer, scientist, etc., and nurse, teacher, librarian, etc.) and two sets of attribute words A, B (e.g., man, male, etc. and woman, female, etc.). The null hypothesis is that the two sets of target words are not different when it comes to their similarity to the two sets of attribute words. We test the null hypothesis using a permutation test on the word embeddings, and the resulting effect size is used to quantify how different the two sets are.

2. **SEAT:** the Sentence Encoder Association Test (May et al., 2019a) was proposed as a contextual version of the popular metric WEAT. As WEAT was computed on static word embedding, in SEAT they proposed using “semantically-bleached” templates such as “This is [target]”, where the target word of interest is planted in the template, to get its word embedding in contextual language models. Thus, we only consider “semantically-bleached” templates to be appropriate as a dataset for SEAT.

3. **CEAT:** Contextualized Embedding Association Test (Guo and Caliskan, 2021) was proposed as another contextual alternative to WEAT. Here, instead of using templates to get the word of interest, for each word a large number of embeddings is collected from a corpus of text, where the word appears many times. WEAT’s effect size is then computed many times, with different embeddings each time, and a combined effect size is then calculated on it. As already mentioned by the original authors, even with only 2 contextual embeddings collected per word in the WEAT stimuli, and each set of X, Y, A, B having only 5 stimuli, $2^{5 \cdot 4}$ possible combinations can be used to compute effect sizes.

4. **Probe:** The entire example, or a specific word in the text, is probed for gender. A classifier is trained to learn the gender from a representation of the word or the text as extracted from a model. This can be done on examples where there is some gender labeling (for instance, the gender of the person discussed in a biography text) or when the text contains some target words, with gender context. Such target words could be “nurse” for female and “doctor” for male. Usually, the word probe refers to a classifier from the family of multilayer preceptron

classifiers, linear classifiers included. The accuracy achieved by the probe is often used as a measure of how much gender information in embedded in the representations, but there are some weaknesses with using accuracy, such as memorization and other issues (Hewitt and Liang, 2019; Belinkov, 2021), and so MDL Probing is proposed as an alternative (Voita and Titov, 2020), and the metric used is compression rate. Higher compression indicates more gender information in the representation.

5. **Cluster:** It is possible to cluster the word embeddings or representations of the examples and perform an analysis using the gender labels just like in probing.

6. **Nearest Neighbors:** As with probing, the examples and word representations can be classified using a nearest neighbor model, or an analysis can be done using nearest neighbors of word embeddings as done by Gonen and Goldberg (2019).

7. **Gender Space:** in the static embeddings regime, Bolukbasi et al. (2016) proposed to identify gender bias in word representations by computing the direction between representations of male and female word pairs such as "he" and "she". They then computed PCA to find the gender direction. Basta et al. (2021) extended the idea to contextual embeddings by using multiple representations for each word, by sampling sentences that contain these words from a large corpus. Zhao et al. (2019) performed the same technique on a different dataset. They then observed the percentage of variance explained in the first principal component, and this measure plays as a bias metric. The principal components can then be further used for a visual qualitative analysis by projecting word embeddings on the component space.

8. **Cos:** in static word embeddings (Bolukbasi et al., 2016), this was computed as the mean cosine similarity between neutral words which might have some stereotype such as "doctor" or "nurse", and the gender space. Basta et al. (2021) computed it on profession words using extracted embeddings from a large corpus.

B Statistical Fairness Metrics

This section describes statistical metrics that are representative of many other fairness metrics that have been proposed in the field. *separation* and *sufficiency* fall under the definition of "extrinsic performance", specifically "gap (Gender)" while *independence* falls under the definition of "extrinsic prediction", specifically "pred gap". Various numbers are generated by these metrics that describe differences between two distributions as measured by Kullbeck-Liebr divergence. We sum all the numbers to quantify bias in a single number.

Let R be a model's prediction, G the protected attribute of gender, and Y the golden labels.

Independence requires that the model's predictions are independent of the gender. Formally:

$P(R = r|Z = F) = P(R = r|Z = M)$

It is measured by the distributional difference between $P(R = r)$ and $P(R = r|Z = z)\ \forall z \in \{M, F\}$.

Separation requires that the model's predictions are independent of the gender *given the label*. Formally:

$P(R = r|Y = y, G = F) = (R = r|Y = y, G = M) \forall y \in \mathcal{Y}$

It is measured by the distributional difference between $P(R = r|Y = y, Z = z)$ and $P(R = r|Y = y) \forall y \in \mathcal{Y}, \forall z \in \{M, F\}$

Sufficiency requires that the distribution of the gold labels is independent of the model's predictions *given the gender*. Formally:

$P(Y = y|R = r, G = F) = P(Y = y|R = r, G = M)$

It is measured by the distributional difference between $P(Y = y|R = r, Z = z)$ and $P(Y = y|R = r) \forall y \in \mathcal{Y}, \forall z \in \{M, F\}$

C Bias in Bios experiments

C.1 Implementation details

In this section we describe the metrics that were measured in the experiments on Bias in Bios, following Orgad et al. (2022).

Performance gap metrics. The standard measure for this task (De-Arteaga et al., 2019) is the True Positive Rate (TPR) gap between male and female examples, for each profession p:

$$TPR_p = TPR_{p_F} - TPR_{p_M}$$

and then compute the Pearson correlation between each TPR_p and the percentage of females in the training set with the profession p. The result is a single number in the range of 0 to 1, with a higher value indicating greater bias. We measure the Pearson correlations of TPR_p, as well as of the False Positive Rate (FPR) and the Precision gaps. In addition, we sum all the gaps in the profession set P, thereby quantifying the absolute bias and not only the correlations, for example, for the TPR gaps: $\sum_{p \in P} TPR_p$.

Statistical fairness metrics. We also measured three statistical metrics (Barocas et al., 2019), relating to several bias concepts: Separation, Sufficiency and Independence. A greater value means more bias. Detailed information on these metrics can be found in Appendix B.

C.2 Correlations between extrinsic and intrinsic metrics when measured on different datasets

Table 3 present the full results of our correlation tests, when intrinsic metrics was measured on a different dataset (Winobias) than the extrinsic metric (Bias in Bios). For all metrics, there is no correlation when we measured the intrinsic metric with a different dataset, although many of the metrics did correlate with the intrinsic metrics when measured on the same dataset as is originally done in Orgad et al..

C.3 Statistics of the Dataset Before Balancing

Table 4 presents how the professions in Bias in Bios dataset (De-Arteaga et al., 2019) are distributed, per gender. The gender was induced by the pronouns used to describe the person in the biography, thus it is likely the self-identified gender of the person described in it.

C.4 Analysis of Precision (Pearson) Metric Behavior

This section discusses the results on the precision (Pearson) metric. This metric is measured as the Pearson correlation between the (i) percentage of women in the training set per profession (ii) precision gap between female examples and male examples in this profession (computed as $\text{precision}_P^F - \text{precision}_P^M$ per profession P).

We observed a strong negative value for the metric when testing the model on a balanced test-set, but a positive value when testing on a test-set with the same distribution as the training-set (which is unbalanced). On an imbalanced test-set, the distribution is similar to the training-set and thus, for female examples, the precision is higher in a profession that is biased in favor of females (meaning, as the percentage of females in the training-set is higher). The same logic applies to the precision of males in professions skewed toward men. Therefore, the correlation is positive. Conversely, if the test-set is balanced, the model's bias hurts the model's precision: as the profession is more biased, for instance the profession of "nurse", for which more than 90% of the training examples are female, the model tends to assign females to the profession more often than necessary, hurting the precision. Figure 5 demonstrate this point: on the imbalanced dataset (Plot 5a), precision for females grows as the percentage of females is larger and precision for males reduces as the percentage of females is larger, and for a balanced dataset (Plot 5b) the trend is reversed. For these reasons, we get a positive correlation with the precision gap on an imbalanced dataset and a negative correlation on a balanced dataset.